

Challenge Problems, Research Readings, and Board Presentations

Discrete Mathematics for Computer Science

Mathematics 4

Introduction

There are two kinds of students in the world of computing.

The first kind learns to use tools. The second kind learns to *build the mind* that can create new tools.

The first kind survives the present. The second kind shapes the future.

This document is for those who want to belong to the second category.

We are living through a civilizational shift. Artificial Intelligence is not just another technology wave. It is a force that is rearranging the value of human skill itself. In such a world, routine coding will get cheaper. Mechanical implementation will get cheaper. Repetitive software labour will get cheaper. The person who only knows how to write yesterday's code will become what the old economy once made out of millions of human beings: a replaceable operator in a giant system.

Tomorrow's world will not reward those who merely *execute*. It will reward those who can *think, model, prove, abstract, present, and create structure out of chaos*.

That is why this initiative matters.

These challenge problems and research-paper presentations are not an “extra activity.” They are part of the deeper work of building your mathematical brain. They are training in clarity. Training in rigor. Training in standing at a board with a marker and making an idea move from your head into the minds of others. Training in reading difficult things without running away. Training in patience with abstraction. Training in intellectual courage.

You must understand this clearly: **serious mathematics is not about memorizing formulas; it is about upgrading the architecture of your mind.**

When Leslie Lamport writes with precision, when Alan Turing defines computability, when Claude Shannon turns communication into mathematics, when Gödel shows the limits of formal systems, when Shamir turns secrecy into structure — they are not merely “solving questions.” They are creating new ways for humanity to think.

And that is exactly the kind of muscle we want to build here.

This is why you must read papers. This is why you must struggle with hard problems. This is why you must present on the board. This is why you must learn to stand in front of others and say: *Here is the idea. Here is the structure. Here is the proof. Here is why it matters.*

Introduction

A strong mind is not built by passive consumption. A strong mind is built by wrestling with difficult ideas until they become part of your nervous system.

If you do this seriously, this activity will make you:

- more rigorous,
- more articulate,
- more fearless in front of abstraction,
- more capable of reading original sources,
- more comfortable with deep technical ideas,
- and far more prepared for the world that is coming.

The future will belong to those who can think beyond the interface.

Let this be your training ground.

1. Format of the Initiative

From this month onward until the end-semester examinations, we will run a weekly **Challenge Problem and Research Reading Presentation Series**.

Time: Every Friday, 5:30 PM

Fallback Rule: If Friday is a holiday, the session will be conducted on Thursday at the same time.

Students may choose **any one challenge problem / research reading / mathematical idea** from the curated list below and prepare:

- a board presentation,
- a clean explanation of the underlying mathematics,
- and, wherever relevant, a discussion of why the topic matters to computer science.

2. Who Should Participate?

2.1. Presenters

Any student who wishes to present may choose one topic, prepare deeply, and present it using only:

- board,
- markers,
- mathematical clarity,
- and intellectual honesty.

2.2. Audience

Other students are **strongly advised** to attend these sessions and observe:

- how peers think,
- how they structure solutions,
- how they communicate difficult ideas,
- how they handle questions,
- and how real mathematics is done in public.

Watching a serious presentation is itself part of mathematical training.

3. Objectives of This Series

- To develop mathematical courage.
- To strengthen proof-writing and proof-speaking ability.
- To cultivate the habit of reading original papers and classical sources.
- To connect discrete mathematics with computer science, cryptography, logic, coding theory, algorithms, complexity, and formal systems.
- To train students to communicate technical ideas with precision.
- To build a culture where difficult ideas are not avoided, but pursued.

4. How to Prepare a Presentation

Each presentation should aim to answer the following questions:

1. What is the problem or idea?
2. What is the historical or intellectual context?
3. What are the central definitions?
4. What is the main theorem / result / construction?
5. What is the proof idea or mechanism?
6. Why does this matter for mathematics or computer science?
7. What are the limitations, open questions, or modern connections?

5. Presentation Expectations

- Do not merely read slides or notes.
- Use the board to build the idea step by step.
- Define all important terms clearly.
- Do not hide behind jargon.
- If you do not understand something, say so honestly.
- If you simplify something, say exactly what has been simplified.
- Focus on insight, not performance.
- A beautiful presentation is one in which the audience leaves with a new structure in the mind.

6. Curated Topic List

6.1. Information, Logic, Computation, and Foundations

Topic	Resource / Link
Claude Shannon's <i>A Mathematical Theory of Communication</i>	https://people.math.harvard.edu/~ctm/home/text/others/shannon/entropy/entropy.pdf
Gödel's Incompleteness Theorem	Hal Prince book — https://books.google.co.in/books/about/The_Annotated_G%C3%B6del.html?id=QxZWzWEACAAJ&source=kp_book_description&redir_esc=y
<i>On Formally Undecidable Propositions of Principia Mathematica and Related Systems</i>	https://monoskop.org/images/9/93/Kurt_G%C3%B6del_On_Formally_Undecidable_Propositions_of_Principia_Mathematica_and_Related_Systems_1992.pdf
Alan Turing's <i>On Computable Numbers</i>	• Paper Link - https://www.cs.virginia.edu/~robins/Turing_Paper_1936.pdf • YouTube Student Presentation - https://www.youtube.com/watch?v=PMdN3N9LhB0 • YouTube Video To Learn More Concept https://www.youtube.com/watch?v=kmAc1nDizu0
Propositions as Types	1. https://homepages.inf.ed.ac.uk/wadler/papers/propositions-as-types/propositions-as-types.pdf 2. https://youtu.be/ixME1qszwZg?si=KXQ0SpbsGGwpfdut
Automated Theorem Proving using LEAN - Introduction	https://www.youtube.com/watch?v=7Ko3QZx4saU&t=77s

Generating Compiler Optimization from Proofs	1. Paper Link -https://www.cs.cornell.edu/~ross/publications/proofgen/proofgen_tate_popl10_tr.pdf 2. YouTube Link -https://www.youtube.com/watch?v=c6NB_aPmsEw
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6.2. Algebra, Number Theory, and Symmetry

Topic	Resource / Link
Abel's Impossibility Theorem	1. https://mathcenter.oxford.emory.edu/site/math108/abel/abels_impossibility_proof.pdf 2. https://web.williams.edu/Mathematics/lg5/ArnoldQuintic.pdf
Counting Problems	• A regular polygon has 97 vertices, labelled in clockwise order as $0, 1, 2, \dots, 96.$ Each vertex can be colored using one of two colors: black or white. Two colorings are considered the same if one can be transformed into the other by a rotation of the polygon. That is, rotating the polygon by any multiple of $\frac{360^\circ}{97}$ does not create a new coloring. Answer the following: 1. Describe the problem of counting all distinct vertex colorings of the 97-gon up to rotation. 2. Determine how many such distinct colorings exist. • Codeforces Problem : https://codeforces.com/contest/1065/problem/E • Problem B-Buildings GCPC-2017 https://shorturl.at/ZuP9i
Adi Shamir — <i>How to Share a Secret</i>	https://web.mit.edu/6.857/OldStuff/Fall03/ref/Shamir-HowToShareASecret.pdf

PRIMES in P — <i>The Prime Facts: From Euclid to AKS</i> by Scott Aaronson	https://www.scottaaronson.com/writings/prime.pdf
15 Puzzle and Rubik's Cube through Group Theory	https://kconrad.math.uconn.edu/blurbs/grouptheory/15puzzle.pdf
Explain $BB(5)$ using Scott Aaronson's big numbers essay	https://www.scottaaronson.com/writings/bignumbers.html

6.3. Computer Science, Coding, Cryptography, and Systems

Topic	Resource / Link
ECC and the Bitcoin paper	https://bitcoin.org/bitcoin.pdf
Generating Compiler Optimization from Proofs	• https://www.cs.cornell.edu/~ross/publications/proofgen/proofgen_tate_popl10_tr.pdf • YouTube Link of Student Presentation https://www.youtube.com/watch?v=c6NB_aPmsEw
Nand Game — building a computer from basic components, complete all levels	https://nandgame.com/
Sample Presentation Link	https://www.youtube.com/watch?v=c6NB_aPmsEw

6.4. Game Theory and Decision Making

Topic	Resource / Link
Basic Introduction to Game Theory	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1968579
The Bargaining Problem	https://www.haverford.edu/sites/default/files/Nash1950.pdf
Sample Presentation Link	1. Basic Introduction to Game Theory: https://www.youtube.com/watch?v=UMhWUDxNqIw 2. The Bargaining Problem: https://www.youtube.com/watch?v=zPHAp00MwGg

6.5. Data, Geometry, and Advanced Structures

Topic	Resource / Link
Sheaf Theory as a Foundation for Heterogeneous Data Fusion	https://digitalcommons.usu.edu/cgi/viewcontent.cgi?params=/context/etd/article/8483/&path_info=2018_Mansourbeigi_Seyed.pdf

7. Video Resources

The following videos may also be chosen as the basis for a board presentation, provided the student goes beyond summary and extracts the mathematics carefully.

- <https://www.youtube.com/watch?v=g5yRQ77borA>
- <https://www.youtube.com/watch?v=kmAc1nDizu0>
- <https://www.youtube.com/watch?v=PMdN3N9LhB0>

8. Suggested Presentation Structure

A good presentation may follow the structure below:

1. **Title and context**
2. **Historical motivation**
3. **Definitions**
4. **Central theorem / construction / problem**
5. **Main proof idea or mechanism**
6. **One worked example**
7. **Connection to computer science**
8. **What makes this idea profound**

9. Evaluation Philosophy

This is not meant to be a ritual of speaking. This is meant to be a ritual of thinking.

The point is not to impress. The point is to understand and to communicate understanding.

A good presentation is one in which:

- the presenter shows evidence of real struggle,
- definitions are respected,
- the mathematical structure is visible,
- and the audience comes away sharper than before.

10. Why This Matters in the Age of AI

The age of AI will divide people into two categories:

- those who use systems,

- and those who understand structure deeply enough to build, judge, improve, and challenge systems.

When machines become better at shallow production, human value shifts upward toward:

- abstraction,
- proof,
- modeling,
- taste,
- judgment,
- and original thought.

That is why you must not reduce your education to assignment completion or syntactic coding.

The coming world will not ask only:

Can you code?

It will increasingly ask:

Can you think clearly enough to define the problem? Can you model it? Can you reason rigorously? Can you explain it? Can you see what others cannot yet see?

That is what this series is trying to cultivate.

11. Final Note to Students

Do not underestimate what repeated exposure to hard ideas can do to your mind.

A student who reads only summaries remains dependent. A student who reads papers begins to grow. A student who presents difficult mathematics on a board begins to transform.

That transformation is the point.

Come not merely to attend. Come to sharpen the instrument through which all your future work will happen: **your mind.**

Gurpreet Singh